Practice 1 Partial 3: Introduction to Statistical Estimation in RStudio

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General Objective

Learn the concept of **statistical estimation** and apply its types (**point estimation** and **interval estimation**) for data analysis in RStudio.

Introduction

Key Concept: Estimation

Estimation is a statistical process by which, based on data collected from a sample, we calculate values that serve to approximate unknown parameters of a population.

Preliminary Research (Complete before the practice)

Before starting the practice, research and write a brief explanation of the following concepts:

1. **Sample Mean** The sample mean is the average value (or mean value) of a sample of numbers taken from a larger population of numbers, where "population" indicates not number of people but the entirety of relevant data, whether collected or not.
2. **Sample Proportion** Percentage of items in a sample that has a specific characteristics
3. **Population Parameter** In statistics, a population parameter is a number that describes something about an entire group or population. This should not be confused with parameters in other types of math, which refer to values that are held constant for a given mathematical function. Note also that a population parameter is not a statistic, which is data that refers to a sample, or subset, of a given population.
4. **Sample and Population** . In statistics, a population is a set of similar items or events which is of interest for some question or experiment. . A sample refers to a subset, or part, of that population. Statistical studies typically use samples instead of populations because it may be costly, time-consuming, or simply impossible to find or reach out to everyone in a population.

5. **Point Estimation** In statistics, point estimation involves the use of sample data to calculate a single value which is to serve as a “best guess” or “best estimate” of an unknown population parameter (for example, the population mean). More formally, it is the application of a point estimator to the data to obtain a point estimate.
6. **Interval Estimation** In statistics, interval estimation is the use of sample data to estimate an interval of possible values of a parameter of interest. This is in contrast to point estimation, which gives a single value.
7. **Confidence Interval** Confidence intervals are used to estimate the parameter of interest from a sampled data set, commonly the mean or standard deviation. A confidence interval states there is a 100γ% confidence that the parameter of interest is within a lower and upper bound.
8. **Confidence Level (e.g., 95%)** In statistics, a confidence level represents the probability that a calculated confidence interval contains the true value of a population parameter. It’s a measure of certainty that the interval, derived from a sample, accurately reflects the range within which the true population value lies. A higher confidence level indicates a greater degree of certainty, but also results in a wider interval. The confidence level, expressed as a percentage, indicates the frequency with which the confidence interval will contain the true population parameter if the sampling process were repeated many times
9. **t-Test (for estimating the mean when population standard deviation is unknown)** A t-test, also known as a Student’s t-test, is a statistical test used to compare the means of two groups and determine whether there is a significant difference between them. It is used to test statistical hypotheses and assess whether two groups are different or whether one group differs from a known value.
10. **Proportion Test (prop.test in R)** A proportion test is a statistical test used to compare a sample proportion to a population proportion or to compare two sample proportions. It helps determine if the observed differences are statistically significant or likely due to random chance.

These concepts are essential for understanding and successfully completing the practice.

Types of Estimation

Point Estimation:

Provides a single numerical value as an approximation of the population parameter.

Example:
“The sample mean is 40 hours.”

Interval Estimation:

Provides a range of values (confidence interval) that, with a certain confidence level (e.g., 95%), contains the true population parameter.

Example:
“The population mean lies between 38 and 42 hours with 95% confidence.”

Normal distribution formula

$f(x) = (1 / (\sigma * \sqrt{(2\pi)})) * e^{(-(x - \mu)^2 / (2\sigma^2))}$

Practice Development

Scenario:

A company wants to know the average number of hours employees work per week and the proportion of employees who work more than 45 hours. Since surveying all employees is not feasible, a sample is taken.

Step 1: Data Simulation

```
# Generating a random sample of 30 employees
set.seed(123)  # Seed for reproducibility
work_hours <- rnorm(30, mean = 40, sd = 5)

# Display first observations
head(work_hours)

## [1] 37.19762 38.84911 47.79354 40.35254 40.64644 48.57532
```

Step 2: Point Estimation of the Mean

Calculate the Work_hours mean:

```
mean = sum(work_hours)/length(work_hours)
cat("The mean value of the work hours is of aproximetly: ", mean,"hours")

## The mean value of the work hours is of aproximetly:  39.76448 hours

#We can use also mean(work_hour)
```

Step 3: Interval Estimation for the Mean

Set 95% Confidence

```r
# Confidence interval for the mean
value <- 0.95
t.test(work_hours, conf.level = value)
```

```
##
##  One Sample t-test
##
## data:  work_hours
## t = 44.402, df = 29, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  37.93287 41.59610
## sample estimates:
## mean of x
##  39.76448
```

Step 4: Point and Interval Estimation for Proportion

Now estimate the proportion of employees who work more than 45 hours per week:

```r
# Number of employees working more than 45 hours
working_more_45 <- sum(work_hours >= 45)
n <- length(work_hours)

# Point estimation of the proportion
sample_proportion <- working_more_45 / n
cat("The point estimation of proportion is", sample_proportion)
```

```
## The point estimation of proportion is 0.1666667
```

```r
# Confidence interval for the proportion
prop.test(working_more_45, n, conf.level = 0.95)
```

```
##
##  1-sample proportions test with continuity correction
##
## data:  working_more_45 out of n, null probability 0.5
## X-squared = 12.033, df = 1, p-value = 0.0005226
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
##  0.06303555 0.35451084
## sample estimates:
##         p
## 0.1666667
```

Print all your results in console

Step 5. Answer the following questions based on your results:

a) What was the sample mean obtained? 0.1666667
b) What was the confidence interval for the mean? 95%

c) What was the sample proportion of employees working more than 45 hours? approximately 1/6 of population
d) What was the confidence interval for the proportion? 95%
e) Why is the confidence level important in these results? To ensure (statistically) the probability that the values we are searching for are in the range of numbers (being 100% the minimum and the maximum of the population).

Final conclusion

The sample shows that 1/6 of people work more than 45h. However, because we used a sample, we can not be 100% sure that it will be exactly 1/6 out od all population. That’s why we used a range that includes the true population. This range goes from 6% to 35% and we have a 95% of certainty that the true value is in there. This indicates that our estimate has considerable variability and it would be advisable to collect more data for great accuracy.